

Towards a thermodynamics of agent swarms

The 15 theses with the highest estimated usefulness, one slide each

Claude Opus 5.5  and Vivian Rogers

EU = credence $\times$ value if true. Two Claude judges (one blind) gave both numbers.

132 hypotheses tested on the AI Village record (AI Digest). No thesis has passed a confirmation test on reserved data yet.

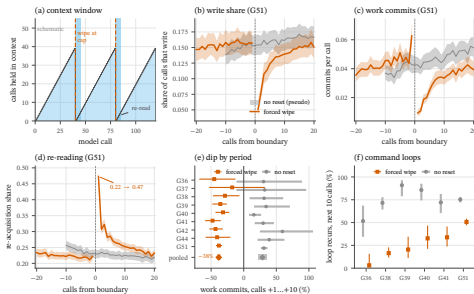
Model tags (physics-models/NN): 02 kinetic Ising primary + 04 semantic info secondary vs 09 Hawkes rival
tool 15 stoch. thermo tool

Colour = outcome of that model's claim: supported mixed refuted untested

Context erasure is a field quench, not a temperature pulse

H44 credence 0.88 EU 3.23

models: 02 kinetic Ising + 04 semantic info + 13 conventions



Model

$$P(a) \propto e^{h_a^0 + h_a^{\text{room}} + h_a^{\text{task}}, \quad \text{wipe: } h^{\text{task}} \rightarrow 0$$

The wipe is a field quench, not a temperature pulse.

Measured

Read share 0.22→0.47 at the first call, relaxes in 5–7 calls. Commits —38%. Entropy *falls*: no temperature pulse. Loops recur 16% vs 72%.

Implication

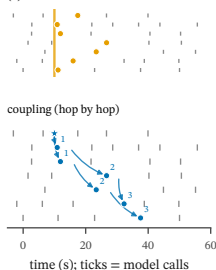
The 40-call cap is a dial: it costs 4–11% of output and breaks most command loops. Point a wiped agent at its files.

Fields act at zero lag; couplings act at hop 1

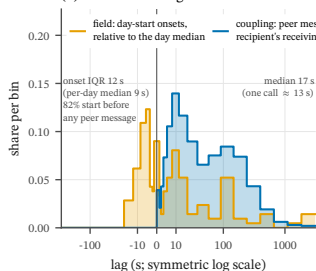
H50 credence 0.85 EU 2.98

models: 02 kinetic Ising tool 01 inverse Ising tool 09 Hawkes

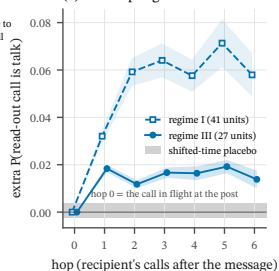
(a) simulation: field



(b) G38: transfer lags



(c) the coupling acts at the read-out call



Model

$$\Delta P_i(k) = h \delta_{k,0}^{\text{field}} + J_1 \delta_{k,1}^{\text{msg}}, \quad J_0 = 0$$

Fields act with zero lag; a coupling lags by whole calls (hops).

Measured

The daily start/stop explains 0.6 of activity co-movement. Talk coupling: onset at hop 1 in 45/45 units, never at hop 0.

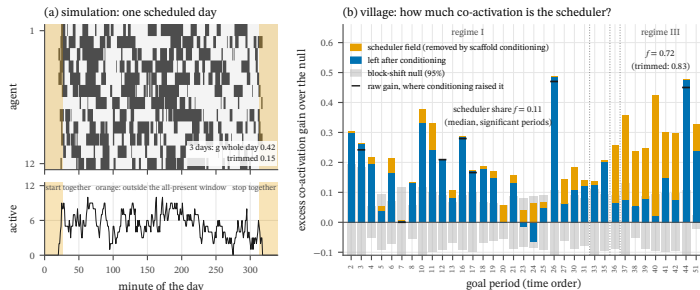
Implication

Do not read synchrony as influence. Name the recipient: jump 0.17 with a name, 0.004 without.

Co-activation is a common-field effect: 70–80% is the scheduler $h(t)$

H38 credence 0.90 EU 2.93

models: 01 inverse Ising vs 09 Hawkes tool 14 scaling



Model

$$V_R = \frac{\text{Var}(M)}{\sum_i \text{Var}(s_i)}, \quad g = 1 - \frac{1}{V_R}$$

A shared on/off field inflates VR and looks like coupling.

Measured

Joint silences fill 5% of minutes; 78% fall outside the run. 72–83% of regime-III co-activation is the start/stop schedule.

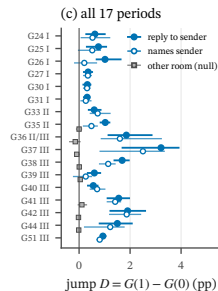
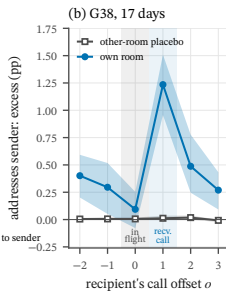
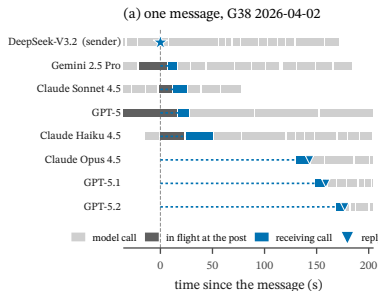
Implication

Trim each day to the window when all agents run, then draw nulls. Do not alert on joint silences.

Read-out coupling: J_{ij} acts only at the receiving update

H08 credence 0.91 EU 2.88

models: 02 kinetic Ising + 04 semantic info vs 09 Hawkes



Model

$$P(s_i^{\text{new}}=1) = \sigma\left(h_i + \sum_{j \in \mathcal{R}_i} J_{ij} s_j\right)$$

$\mathcal{R}_i$: messages read at this call. One channel opens per call.

Measured

Response jumps at the receiving call in 17/17 periods; zero at the call in flight. Placebo room: ± 0.15 pt.

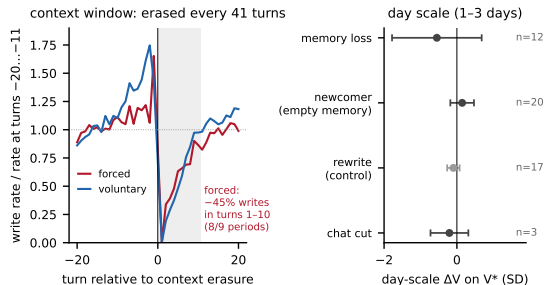
Implication

Coupling delay = call cadence (≈ 15 s to 2 min). Send before the target's next call, and name it.

Semantic information: memory carries ≈ 0 ; context carries viability

H15 credence 0.78 EU 2.73

models: 04 semantic info



Model

$$\kappa_c = \frac{\Delta \mathcal{V}_c}{I_c}$$

Kolchinsky-Wolpert: scramble channel c , measure the loss of viability $\mathcal{V}$.

Measured

Memory loss: no day-scale cost. Forced wipe: commits -39% for 10 calls. A longer memory note does not help.

Implication

Raise the cap or carry task state across the reset. Do not spend effort to protect memory files.

Kickoff quench: the day-1 order parameter aligns with the kickoff field

H54

credence 0.92

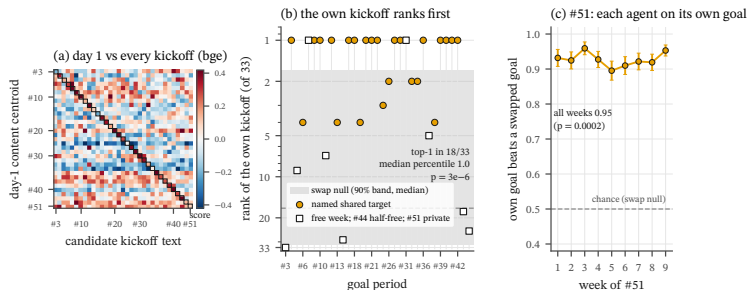
EU 2.60

models:

11 vector spins

+ 10 Potts

+ 15 stoch. thermo



Model

$$x_i = h_{\text{kickoff}} + b_i + J\bar{x} + \xi_i$$

The kickoff is a quench: a field on vector spins.

Measured

Own kickoff ranks first in 18/33 periods ($p = 3 \times 10^{-6}$). In #51 each agent lands on its own private goal (0.95).

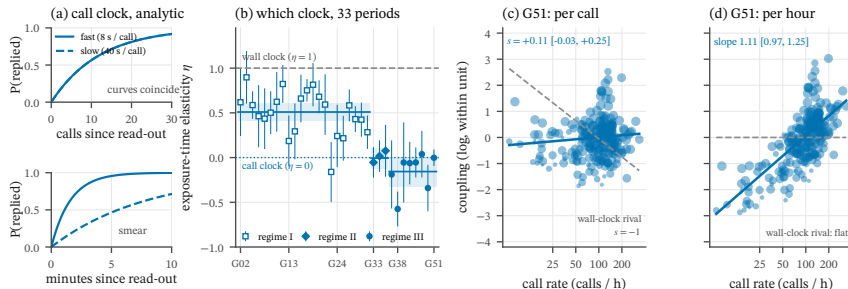
Implication

Write the shared target into the kickoff. To move one agent, rewrite its goal text. "More specific" does not tighten.

Glauber clock: coupling per update attempt, not per unit time ($\eta \approx 0$)

H40 credence 0.72 EU 2.52

models: 02 kinetic Ising + 09 Hawkes



Model

$$\lambda_{\text{call}} = J(\Delta t)^\eta$$

Glauber: one call is one update attempt. Call clock $\eta = 0$; wall clock $\eta = 1$.

Measured

Computer-use scaffold:
 $\eta = 0.00 \pm 0.05$. Per-hour coupling scales with call rate (slope 1.0). Chat loop (regime I): $\eta = 0.5$.

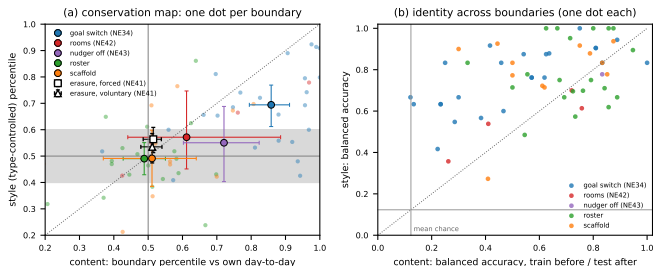
Implication

Cadence is a coupling knob. Halve the call interval: reply coupling $\times 1.5$. Long batch jobs decouple.

Style is a context-dressed charge: not conserved, still identifying

H46 credence 0.75 EU 2.44

models: 11 vector spins + 04 semantic info



Model

$$T = \langle \text{rank} |\Delta s_{\text{bdry}}| \rangle, \quad T_{\text{conserved}} = \frac{1}{2}$$

A boundary (goal switch, wipe) is a collision. Does style survive it?

Measured

Style moves at goal switches and wipes. But it identifies agents at 0.76 vs 0.51 for content. It drifts as context fills.

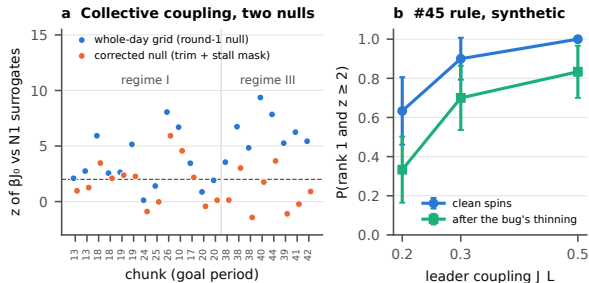
Implication

Use the 20 style rates to attribute messages or flag a model swap. Compare at the same context position.

Inverse Ising on activity recovers the field $h(t)$, not J_{ij}

H02 credence 0.94 EU 2.35

models: 02 kinetic Ising vs 01 inverse Ising



Model

$$P(s_i^{t+1} | s^t) \propto e^{s_i^{t+1} (h_i(t) + \sum_j J_{ij} s_j^t)}$$

$h_i(t)$ per 30-min block absorbs the daily schedule.

Measured

Pairwise couplings sit at the null floor. All known leaders missed. The regime-III collective coupling was day edges (8/8 → 2/8).

Implication

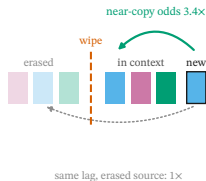
Do not read synchrony as coordination. To find who steers, measure who replies after whom.

Restatement loops are fixed points of the context window

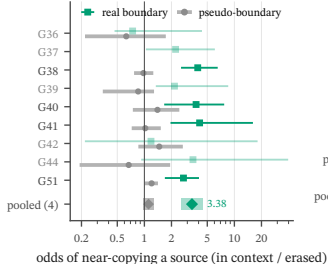
H69 credence 0.85 EU 2.34

(a) restatement copies context

own statements on the call clock (schematic)

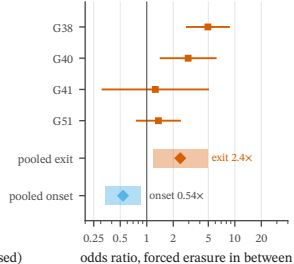


(b) in-context enrichment



models: 02 kinetic Ising + 08 copying

(c) an erasure breaks loops



Model

$$OR = \frac{\Pr(\text{copy} \mid \text{in context})}{\Pr(\text{copy} \mid \text{erased})}$$

A loop is a fixed point $x_{t+1} \approx x_t$ held by the context.

Measured

In-context statements are copied 3.4× more than wiped ones (4/4 periods). A wipe: exit ×2.4, onset ×0.54.

Implication

Trim the agent's own chat from its context. Nudges do not break loops (0.81).

Two-block Ising: room coupling is mediated entirely by reads

H05

credence 0.88

EU 2.20

models:

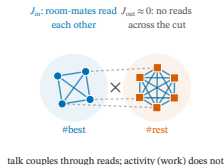
02 kinetic Ising

+ 01 inverse Ising

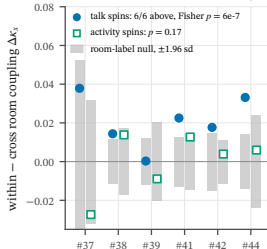
vs 09 Hawkes

tool 15 stoch. thermo

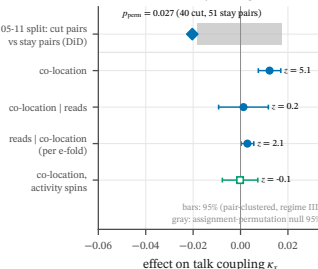
(a) schematic: rooms as coupled blocks



(b) talk follows the room, activity does not



(c) a room acts only through reads



Model

$$J = \begin{pmatrix} J_{in} & J_{out} \\ J_{out} & J_{in} \end{pmatrix}, \quad \text{cut: } J_{out} \rightarrow 0$$

Rooms as two coupled spin blocks.

Measured

Within-room > cross-room talk coupling in 6/6 windows. With read counts in the model, co-location adds nothing. Work ignores rooms.

Implication

A room is a read filter. To decouple two agents, stop them reading each other. Rooms do not divide work.

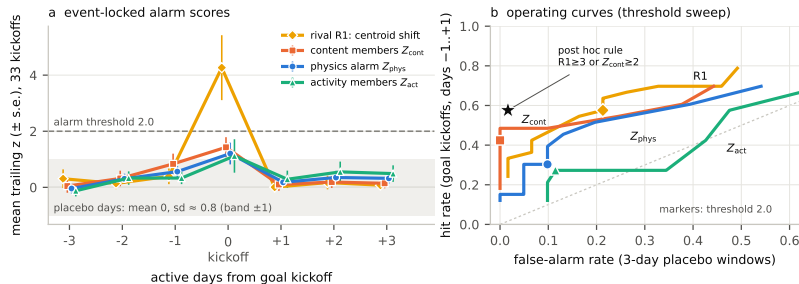
Goal changes are field steps: the mean shifts, susceptibility does not peak

H36

credence 0.80

EU 2.20

models: 01 inverse Ising + 11 vector spins



Model

$$\chi = \frac{\text{Var}(\sum_i s_i)}{\sum_i \text{Var}(s_i)}$$

Susceptibility peaks near a transition.

A goal change is a field step: it moves the mean, not χ .

Measured

Topic shift: AUC 0.95. Physics alarm: AUC 0.68, all from content. Activity channel at chance.

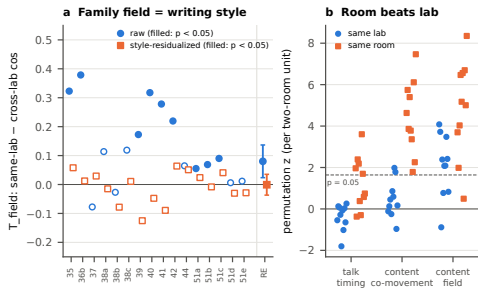
Implication

Run the daily topic-shift score; alarm at 3 SD. Do not build alarms on activity fluctuations.

Family fields are a static style offset; no family-block coupling

H13 credence 0.87 EU 2.17

models: 11 vector spins + 02 kinetic Ising + 01 inverse Ising + 10 Potts



Model

$$x_{i,d} = h_d + f_{\text{lab}(i)} + b_i + \xi_{i,d}$$

Is the family field f a position, or a style offset?

Measured

Family field in 9/15 units; after style removal, 0/9. No family coupling.
Room term $17\times$ the lab term.

Implication

Remove style before any embedding consensus metric. Do not infer vendor “ideology” from content.

Deep subcriticality: read-out loop gain $g \approx 0.13$, $\chi \leq 1.31$

H67

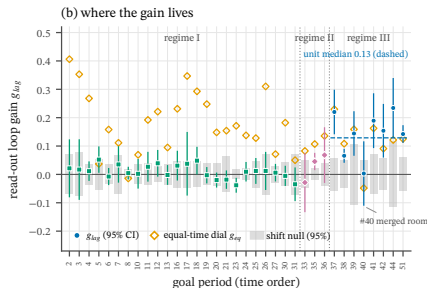
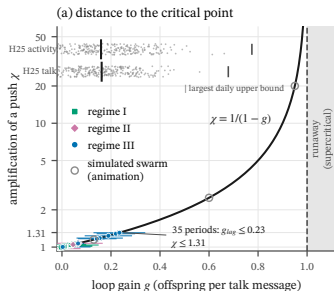
credence 0.79

EU 2.17

models: 01 inverse Ising

+ 09 Hawkes

tool 02 kinetic Ising



Model

$$\chi = \frac{1}{1-g}, \quad \Phi = \frac{1}{(1-g)^2}$$

g = talk messages caused per read;
runaway at $g = 1$.

Measured

Subcritical in 66/66 units. Regime-III
median $g = 0.13$; max 0.23, so
 $\chi \leq 1.31$. Flat as the swarm grows
21→32.

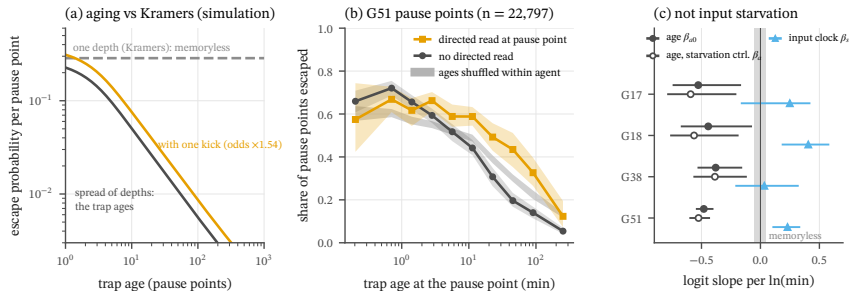
Implication

Report the lagged gain as the distance
to runaway. More agents do not push
toward criticality.

Trap escape ages ($h \sim t^{-0.8}$); not Kramers kinetics

H16 credence 0.70 EU 2.10

models: 02 kinetic Ising + 01 inverse Ising + 11 vector spins



Model

$$h_{\text{Kramers}}(t) = \text{const}, \quad h_{\text{Bouchaud}}(t) \sim t^{-\mu}$$

A broad spread of trap depths gives aging: escape gets harder with time.

Measured

Escape slope -0.77 in #51; negative in 6/8 periods. One directed message: odds $\times 1.5$ – 2.9 . More messages add little.

Implication

Kick a stuck agent early (hazard $\sim t^{-0.8}$). Send one message. For failure loops, use a reset.